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Homoclinic tangle in tokamak divertors

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ABSTRACT

Magnetic field lines are the trajectories of a $1\frac{1}{2}$ degree of freedom Hamiltonians. Plasmas in tokamaks are confined in regions where the magnetic field lines form closed toroidal surfaces. These surfaces are bounded by a separatrix, and outside the separatrix the magnetic field lines and the plasma flow to special regions of the walls called divertors. Both the confinement of the plasma and the feasibility of divertors are sensitive to the behavior of the magnetic field lines near the separatrix in the presence of non-axisymmetric magnetic perturbations. Separatrix manifold forms homoclinic tangle to preserve the symplectic invariant and topological neighborhood as the manifold evolves in canonical time. A scheme is developed based on these two invariants to calculate homoclinic tangle for actual tokamak equilibria that are subjected to non-axisymmetric perturbations. This scheme is used to study homoclinic tangles of a specific tokamak configuration. How non-ideal effects fill in the lobes formed by homoclinic tangles is demonstrated using a radial expansion operator to simulate plasma diffusion. It is found that for sufficiently rapid plasma diffusion the effects of the tangle on plasma footprint on collector plate are washed out.

1. Introduction

The plasma edge in modern tokamaks is defined by a separatrix, which leads magnetic field lines from the edge to special collection regions called divertors [1]. The magnetic field lines just outside the separatrix would enter the divertor region in a circular annulus if the tokamak were exactly axisymmetric, but even small asymmetries to the magnetic field radically changes the behavior of magnetic field lines near the separatrix. This paper shows how a simple mapping procedure can be used to study this effect and in particular the pattern of particle that enter the divertor chamber when these particle arrive as the result of a slow radial diffusion from the plasma region well inside the separatrix. The particles are shown to form lobes which can be sharp curves or broad regions of interception depending on the magnitude of the diffusive transport per toroidal transit of a magnetic field lines. The formation of lobes when a separatrix is perturbed is a venerable subject, which is explained in Section 2 on homoclinic tangles and lobes. After presenting the calculation of homoclinic tangles, the paper will focus on the on the effect of the magnitude of diffusion, or more precisely the random step per toroidal circuit, on filling in the lobes. Filling in the lobes would generally be a good effect because it would spread the heat load.

Magnetic field line trajectories in space are a $1\frac{1}{2}$ degree of freedom Hamiltonian system [2,3]. Hamiltonian mechanics is geometry in phase space, and the phase space has structure of a symplectic manifold [4]. Remarkably the coordinate transformations and topological methods of Hamiltonian mechanics are more natural for the study of the magnetic fields in toroidal plasma confinement devices, such as tokamaks, than for the standard problems of classical mechanics. This is because in classical mechanics, trajectories are generally needed for only a few initial conditions, but in toroidal magnetic confinement, the trajectory of every magnetic field line is needed in the region of space occupied by the plasma. The separatrix surface in divertor tokamaks defines the plasma edge. This surface has a hyperbolic singular point called an X-point. The separatrix is designed to ensure the plasma exhaust from the tokamak goes into a special region called a divertor chamber. An ideal tokamak would be axisymmetric, and the magnetic field lines just outside the separatrix will enter the divertor chamber through a circular annulus. Magnetic asymmetries always exist, and even small asymmetries turn the separatrix into a complicated structure, which has major implications both for tokamak confinement and for divertor design. In particular, non-axisymmetric perturbations cause the magnetic field lines to form a *homoclinic tangle*. After explaining the cause and the features of homoclinic tangles,

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calculations of this tangle for magnetic perturbations in the DIII-D tokamak [5] will be given.

Poincaré discovered homoclinic tangle in his study of three body problem in very simple terms of two gravitating bodies perturbed by a third body of so small a mass that the perturbation from it has negligible effect on the motion of the original bodies. His approach was geometrical in nature. He visualized the behavior of system around the homoclinic fixed point – a saddle point where stable and unstable trajectories intersect. He discovered that the trajectories cross an infinite number of times. He found that each of the two curves can never cross itself but it must fold back on itself an infinite number of times. He was struck by the extreme complexity of motion in phase space [6,7]. At a homoclinic point, four manifolds join together; the two incoming stable separatrix manifolds, and the two outgoing unstable separatrix manifolds. The stable manifold $\mathscr{M}^{\mathcal{S}}$ leading to and the unstable manifold $\mathscr{M}^{\mathcal{U}}$ emanating from the hyperbolic point have extremely irregular behavior. This is because these two manifolds cannot intersect themselves but the unstable manifold $\mathscr{M}^{\mathcal{U}}$ can intersect the stable manifold $\mathscr{M}^{\mathcal{S}}$ at *homoclinic points*. Between each homoclinic point and the hyperbolic fixed point, there are an infinite number of homoclinic points. Thus the stable manifold $\mathscr{M}^{\mathcal{S}}$ and the unstable manifold $\mathscr{M}^{\mathcal{U}}$ form an extremely complex network called the *homoclinic tangle* [8,9]. For manifolds connected to neighboring hyperbolic points, these structures are called *heteroclinic tangle*. In equilibrium, the axisymmetric separatrix manifold is degenerate, and the stable and unstable manifolds coincide.

The magnetic field in toroidal confinement schemes can be expressed as $\vec{B} = \nabla\psi \times \nabla\theta + \nabla\varphi \times \nabla\chi(\theta,\psi,\varphi)$ [2,3] where χ is a poloidal magnetic flux contained inside a magnetic surface and it plays the role of Hamiltonian, θ is a poloidal angle and plays the role of generalized coordinate, ψ is a toroidal magnetic flux inside a surface and plays the role of generalized momentum, and φ is a toroidal angle and plays the role of generalized time. There are three sets of canonical coordinates for magnetic field line trajectories in tokamaks; magnetic, natural, and physical. Physical coordinates are (x, y, φ) where $x = r\cos(\theta)$, $y = r\sin(\theta)$ for $\varphi =$ constant; and r is the radial distance from magnetic axis. Natural coordinates (NCC) are (ψ, θ, φ) where $\psi = B_0 r^2/2$ where B_0 is the magnetic field on magnetic axis, and $\theta = \tan^{-1}(y/x)$ [10]. NCC can be mapped to physical coordinates. Physical coordinates give the position in real physical space. Physical coordinates can be readily mapped to the standard coordinates such as rectangular or cylindrical coordinates. Realistic equilibrium Hamiltonian for field lines in devices such as DIII-D can be accurately and analytically expressed in NCC [11,12]. Magnetic coordinates are averages over magnetic surfaces.

The most efficient way to study the homoclinic tangle of separatrix is to utilize a symplectic map [9], and this is done, for example in [13,14] in cases where the unperturbed plasma is near circular. However, in large modern tokamaks the magnetic surfaces in the edge and the separatrix have a topology that is fundamentally different from that of near-circular plasma. We have developed a scheme to study this fundamentally more complicated case. It is based on the principle that tangles are produced to preserve two invariants – the symplectic invariant and neighborhoods on the separatrix manifold. In our scheme, these two are preserved as the separatrix manifold in phase space evolves forward and backward in canonical time and intersects itself successively in $\varphi = 0$ plane. We integrate the separatrix manifold using maps in NCC. The total Hamiltonian $\chi(\psi,\theta,\varphi) = \chi_0(\psi,\theta) + \chi_1(\psi,\theta,\varphi)$ where $\chi_0(\psi,\theta)$ is the equilibrium poloidal flux and $\chi_1(\psi,\theta,\varphi)$ is the magnetic perturbation. $\chi_0(\psi,\theta)$ for a specific device is a complicated analytic function. The separatrix manifold is made of N points or field lines with $\theta_i = 2\pi i/N$, $i = 1, 2, \ldots, N$. N is large, generally $N = 360$ K or 100 K, and ψ_i is solution of $\chi_0(\psi_i,\theta_i) = \chi_{SEP}$. χ_{SEP} is the equilibrium poloidal flux inside the separatrix. This manifold is compact and closed and preserves the topological neighborhood. We advance this manifold with maps in NCC both forward and backward in the toroidal angle φ. When the forward and backward advanced manifolds intersect in $\varphi = 0$ plane, the preservation of symplectic invariant and topological neighborhoods generate the successive homoclinic tangles. The successive tangles here are not the homoclinic tangle that is created in the limit as $\varphi \to \pm\infty$. The successive tangles can generate complicated electric fields in the edge starting as early as the first intersection, and modify the trajectories of guiding centers of plasma particles. So the plasma particles may not see the homoclinic tangle from the limit $\varphi \to \pm\infty$. Our scheme is different than calculating tangles by following single trajectories in phase space, for example, as in [15] or calculating the trajectories asymptotically in canonical time, for example, as in [16]. Our scheme considers dynamical evolution of a manifold in phase space subject to perturbation and constrained by two topological invariants. This scheme can be used for magnetic confinement configurations such as tokamaks, stellarators, snow flake divertors, etc. Here, we apply this scheme to the DIII-D tokamak [5].

The equilibrium generating function (EGF) in the NCC for magnetic confinement devices can be constructed from the equilibrium Grad–Shafranov solver data, and can realistically and accurately represent the complicated magnetic topology in edge [11,12]. The EGF for the DIII-D edge magnetic geometry is a very complicated. It is a mixed bivariate polynomial in u and v, $\chi_0(\psi,\theta) = \sum_{i=1}^{5} a_i u^i + \sum_{j=1}^{6} b_j v^j + \sum_{i=1,j=1}^{i=5,j=6} c_{ij} u^i v^j$, where $u = \sqrt{(\psi)}\cos(\theta)$ and $v = \sqrt{(\psi)}\sin(\theta)$ [11,12]. It accurately represents the complicated shape of the equilibrium magnetic surfaces in the DIII-D for the shot 115 467 at 3000 ms [11,12]. On the other hand, the simple map [17] has generic magnetic topology of a single-null divertor tokamak; and it has the simplest possible EGF, $\chi_0(\psi,\theta) = u^2 + v^2 + \tilde{b}_3 v^3$ where $\tilde{b}_3 =$ constant [18]. The DIII-D and the simple map both have the same topology – a single null separatrix – but they have vast differences in the shape of equilibrium surfaces. The DIII-D has surfaces with very complicated shape; the simple map has the surfaces with the simplest shape. So the work reported here also sheds some light on what effects the differences in shape has on homoclinic tangles and the chaos in Hamiltonian or near-Hamiltonian systems with the same topology of a single-null separatrix. It gives us a comparison of homoclinic tangle for a very complicated shape of equilibrium surfaces versus the simplest shape for the same topology. We find that the perturbations create very complicated structures in phase space to preserve the two invariants; and as the number of toroidal circuits is increased, tangle becomes extremely complicated; and the simple map is more sensitive to perturbations than the DIII-D.

The model of the perturbation is simplified in two ways: (1) Only a cosine series is used in a Fourier decomposition of the perturbation in the poloidal and toroidal angles, and (2) the radial dependence of the perturbation is ignored. The scale of the radial variation of the various Fourier terms in perturbation is given by the wavenumber defined by their poloidal or toroidal dependences and should not be strong in the separatrix region. When r/m is smaller than the width of scrape-off layer in tokamak, ignoring the radial dependence of the perturbation is well justified. Here r is radial coordinate, and m is the poloidal mode number of the Fourier mode in perturbation. A symplectic, accurate, and realistic calculation of successive homoclinic tangle in a real tokamak having surfaces with complicated shape and in the simple map having surfaces with the simplest shape in edge is done. Then results on how the magnitude of plasma diffusion fills the lobes (see Section 2) and its effects on the topology of the plasma footprint on collector plates will be given. Filling of lobes from diffusion is generally a positive effect since it spreads the heat load on plates.

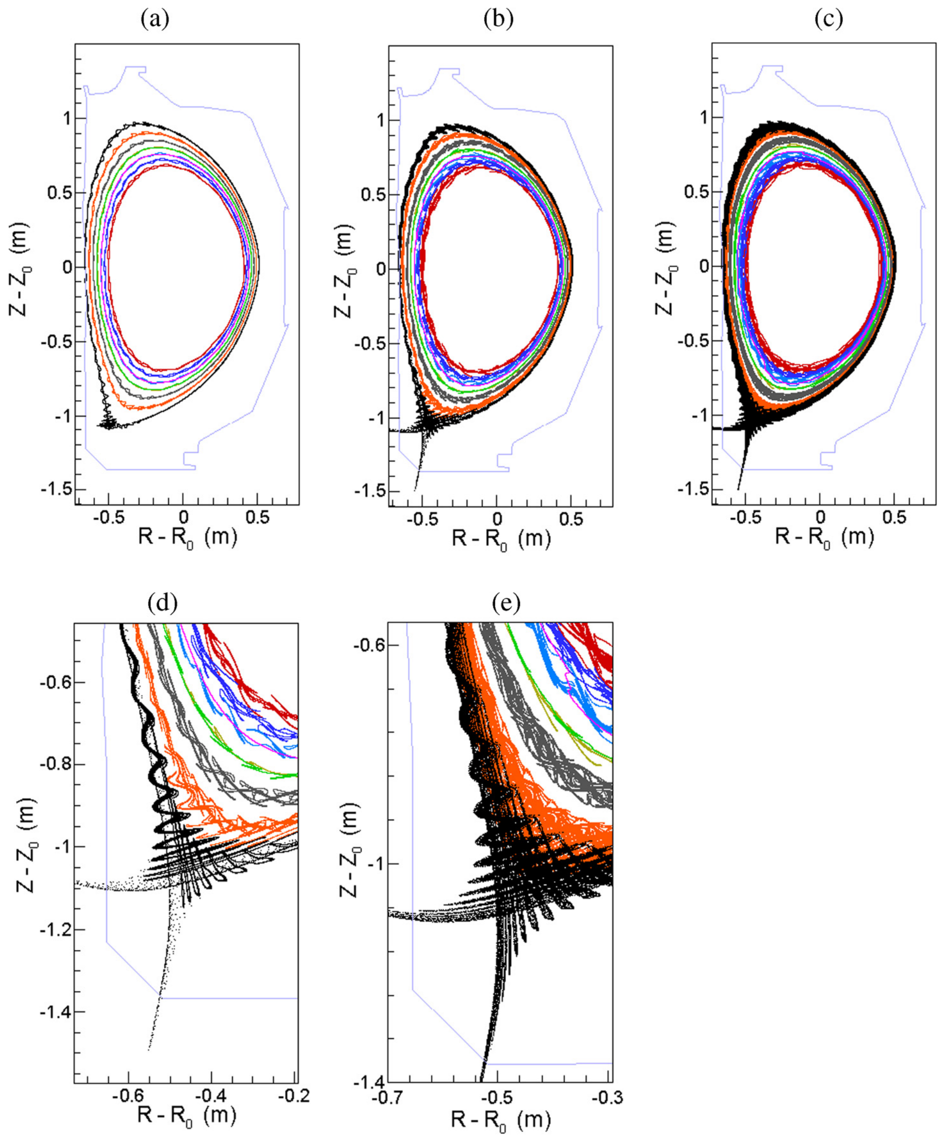

Fig. 1. Homoclinic tangles of the ideal separatrix and heteroclinic tangles of surfaces $\chi = 0.95\chi_{SEP}, 0.9\chi_{SEP}, .., 0.7\chi_{SEP}$ in the DIII-D from modes $(30, 10) + (40, 10)$ with $\delta = 10^{-4}$ in $\varphi = 0$ plane. Each equilibrium surface consists of 360 K points distributed uniformly in poloidal angle θ. (a) after 1 toroidal circuit, (b) 5 circuits, (c) 10 circuits, (d) enlarged view of (b) near the X-point, and (e) enlarged view of (c). χ_{SEP} is poloidal flux inside separatrix. (R, Z) are cylindrical coordinates, and (R_0, Z_0) is the position of magnetic axis.

2. Lobes and homoclinic tangle

Lobes are expected whenever a tokamak separatrix is perturbed to form homoclinic tangle by non-axisymmetric magnetic fields. This is best understood using the mapping of field lines from one toroidal transit to the next. During each transit of a torus, a magnetic field line goes from a point $\vec{x}$ to a point $\vec{x}'$, which actually means from R to R' in major radius and form z to z' in vertical position. This change can be written using a mapping function $\vec{x}' = \vec{T}(\vec{x})$. After two circuits of the torus, a field line goes from a point $\vec{x}$ to $\vec{x}'$, where $\vec{x}' = \vec{T}_2(\vec{x}) \equiv \vec{T}(\vec{T}(\vec{x}))$. The map after n transits of the torus is written as $\vec{x}' = \vec{T}_n(\vec{x})$. The X-point of the tokamak separatrix is a fixed point $\vec{x}_f$ of the map $\vec{T}(\vec{x})$, which means $\vec{x}_f = \vec{T}(\vec{x}_f)$. The separatrix itself in an axisymmetric tokamak is formed by the collection of points $\vec{x}_s$, such that when iterated forward using the map, approach the X-point arbitrarily closely, $\lim_{n\to\infty} \vec{T}(\vec{x}_s) \to \vec{x}_f$. The same separatrix is formed by the collection of points $\vec{x}_u$, such that when iterated backwards, approach the X-point arbitrarily closely, $\lim_{n\to-\infty} \vec{T}(\vec{x}_u) \to \vec{x}_f$. When the magnetic field is perturbed by a non-axisymmetric magnetic field the mapping function $\vec{x}' = \vec{T}(\vec{x})$ retains a fixed point but the curve formed by field lines that under a forward mapping approach the X-point and the curve formed by field lines that under a backward mapping approach the X-point are not the same curve. The first curve is called the stable manifold and the second curve is called the unstable manifold of the map. For small perturbations, the curves that correspond to the stable and the unstable manifolds cross. The point of crossing, or equivalently a point that lies on both manifolds, is called a homoclinic point, $\vec{x}_h$. The definition of the curves that form the stable and unstable manifolds implies that the mapping of any homoclinic point is to another homoclinic point, so $\vec{T}_n(\vec{x}_h)$ is a homoclinic point for arbitrary positive or negative n. The stable and unstable manifolds that pass through two successive homoclinic points bound a region of space, and that boundary can be mapped forward using $\vec{T}(\vec{x})$. The continuity of $\vec{T}(\vec{x})$ implies points enclosed by an arbitrary closed curve cannot escape being enclosed by the mapping of the curve. If the map were area preserving, which is equivalent to $\vec{\nabla} \cdot \vec{T} = 0$, then the area enclosed by a closed boundary would be unchanged by the mapping. The magnetic field is divergence free, and one could

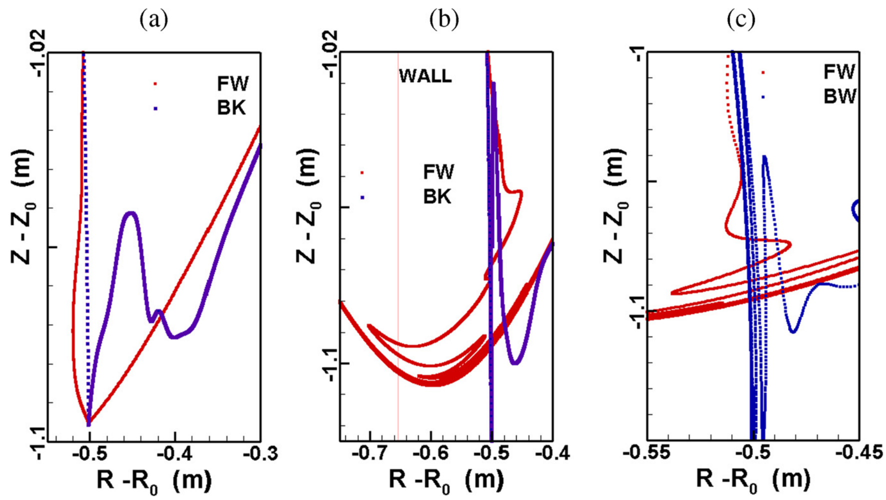

Fig. 2. Homoclinic tangles of the separatrix in the DIII-D from noise and error fields of amplitude 2×10^{-5} close to the X-point in $\varphi = 0$ plane after (a) 1 turn, (b) 3 turns, and (c) 5 turns.

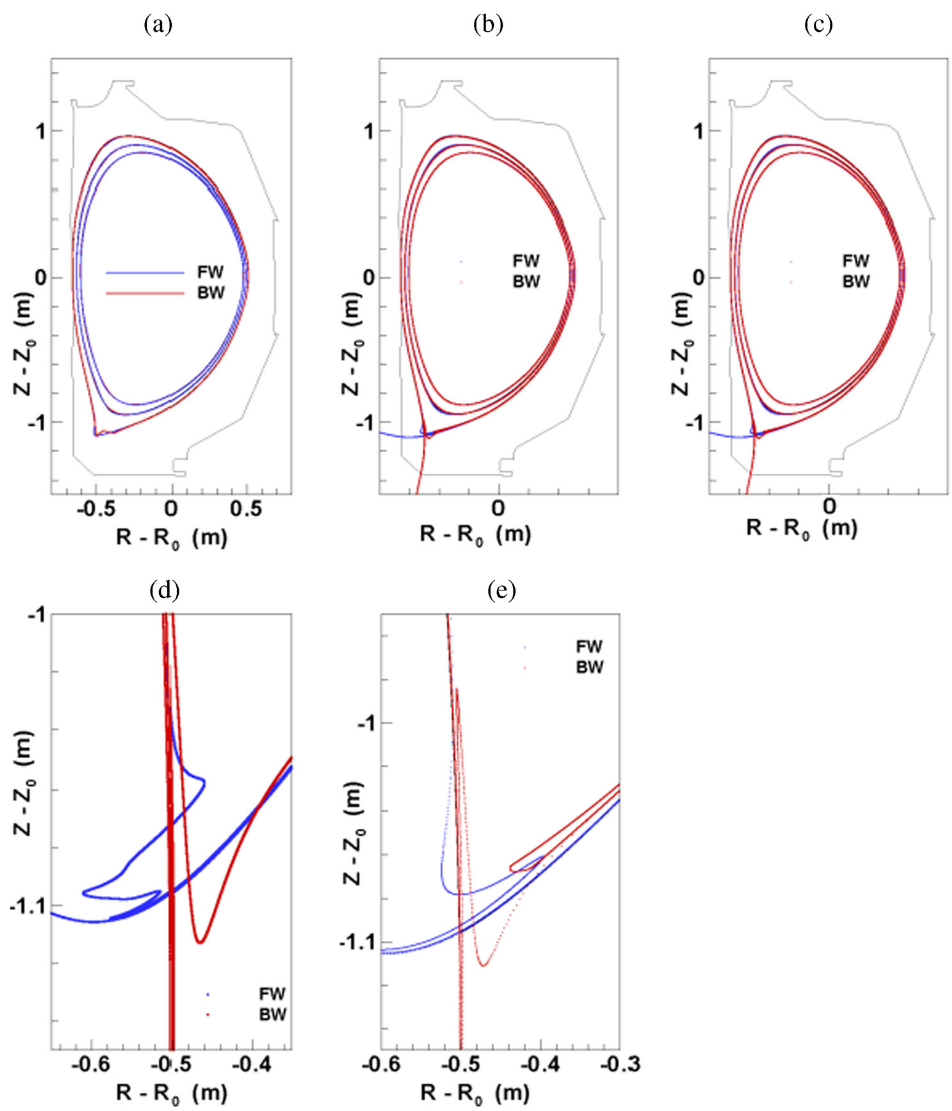

Fig. 3. Tangles of the surfaces $\chi = 0.95\chi_{SEP}$, $0.95\chi_{SEP}$, and $0.9\chi_{SEP}$ in the DIII-D from modes $(3,1) + (4,1)$ with $\delta = 10^{-4}$ in $\varphi = 0$ plane after (a) 1 turn, (b) 5 turns, (c) 10 turns, and enlarged view of homoclinic tangles of separatrix near the X-point after (d) 3 turns, and (e) 5 turns.

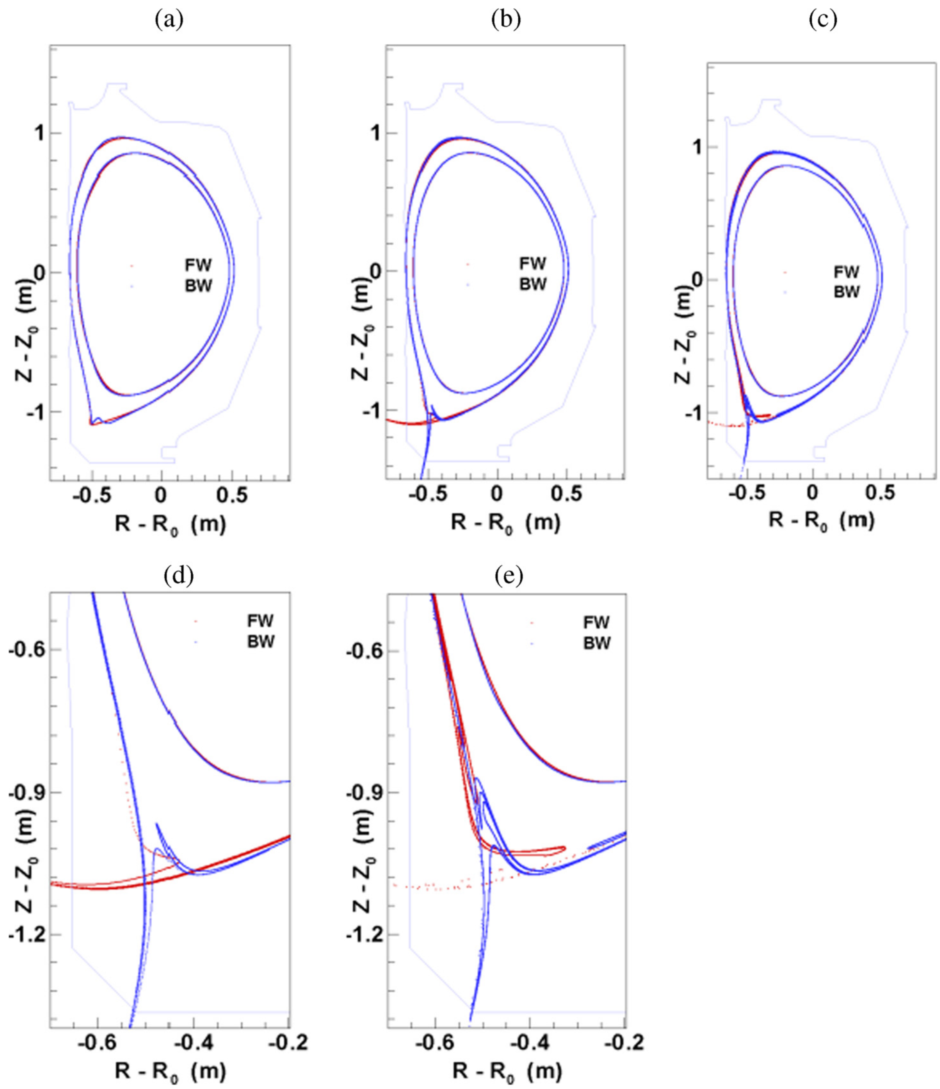

Fig. 4. Tangles of the surfaces $\chi = 0.95\chi_{\mathrm{SEP}}$ and $0.9\chi_{\mathrm{SEP}}$ from modes $(1,1)+(1,-1)$ with $\delta = 6\times 10^{-4}$ in $\varphi = 0$ plane after (a) 1 turn, (b) 5 turns, (c) 10 turns, and enlarged views near the X-point after (d) 5 turns, and (e) 10 turns.

use magnetic flux coordinates and make $\vec{\nabla}\cdot\vec{T} = 0$. Using ordinary spatial coordinates the area enclosed by the boundary times the average toroidal magnetic field is a constant. The term tangle comes from considering what happens as the closed curve defined by the stable and unstable manifolds are mapped using $\vec{T}_n(\vec{x})$. The homoclinic points associated with successive iterations of the map come ever closer to the X-point but never reach the X-point. The implication is successive homoclinic points become ever closer together. Nevertheless, the area enclosed by the stable and manifolds between the two successive homoclinic points must remain essentially constant. The implication is that as $n \to \infty$ the curve that represents the unstable manifold must make wilder and wilder excursions. Similarly, as $n \to -\infty$ the curve that represents the stable manifold must make wild excursions. This will be seen in the results.

3. The method, results, and discussion

The total Hamiltonian is $\chi(\psi,\theta,\varphi) = \chi_0(\psi,\theta) + \chi_1(\psi,\theta,\varphi)$ where $\chi_0(\psi,\theta)$ is the axisymmetric equilibrium flux, and $\chi_1(\psi,\theta,\varphi)$ is the perturbation. The perturbation is generally given by $\chi_1 = \sum_{(m,n)} f_{mn}(\psi)\cos(m\theta - n\varphi + \varphi_{mn})$. m and n are poloidal and toroidal mode numbers of the Fourier components of the magnetic perturbation. We take the modes to be locked with the same amplitude, so $\chi_1(\psi,\theta,\varphi) = \sum_{(m,n)} \delta\cos(m\theta - n\varphi)$. The precise magnetic fields near the edge of a tokamak are not known, but the simplifying assumptions of modes with identical amplitudes that are independent of ψ and locked in phase allow a study of the basic features introduced by the separatrix topology.

The tangles produced by peeling–ballooning modes are represented by including two Fourier components $(m,n) = (30,10) + (40,10)$ with an amplitude $\delta = 10^{-4}$ in Fig. 1. This perturbation represents the type I Edge Localized Modes (ELMs) [19]. In this case, there are a very large number of lobes, and the homoclinic tangle is extremely complicated, especially near the X-point where the lobes are very pronounced.

For internal topological noise and field errors in the DIII-D, we choose Fourier modes $(m,n) = \{(3,1),(4,1),(6,2),(7,2),(8,2),(9,3),(10,3),(11,3)\}$. Internal noise and field errors are ubiquitous in tokamaks. The amplitude δ varies from 8×10^{-6} to 2×10^{-5} [20,21]. We consider the case when $\delta = 2\times 10^{-5}$. For this perturbation, the tangles of separatrix in the DIII-D are shown in Fig. 2. Tangle and lobes from noise and errors are nowhere as complicated as for peeling–ballooning modes; near the X-point the lobes are wide and well-extended radially. We next calculate tangles from the $(3,1)+(4,1)$ modes with amplitude $\delta = 10^{-4}$. These modes are dominant in the edge and results are shown in Fig. 3. In this case, the tangle is quite muted except close to

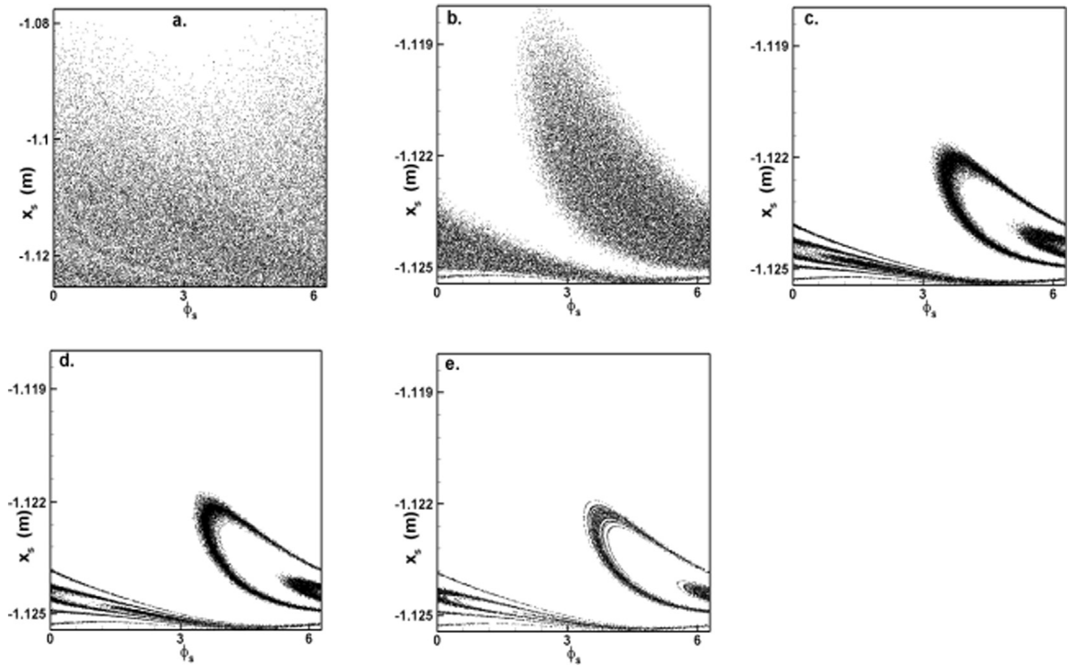

Fig. 5. Footprint of lines starting on a surface 5 cm inside the last good surface for the modes $(3,1)+(4,1)$ and $\delta = 10^{-4}$ for (a) $D = 10^{-2}$, (b) $D = 10^{-3}$, (c) $D = 10^{-4}$, (d) $D = 5\times10^{-5}$, and (e) the footprint of the separatrix for 10 to 1000 circuits.

the X-point where the lobes have a well-discernible and relatively simple structure. The low *mn* perturbation $(1,1)+(1,-1)$ occur naturally in the core, and are estimated to have the amplitude $\delta = 6\times10^{-4}$ [22 and references therein]. Tangles for low mn perturbation are shown in Fig. 4. For the low mn perturbation, again the tangle is quite muted (though not as much as for $(3,1)+(4,1)$ modes) except near the X-point where the lobes have a well-defined structure. In general, we see that the lobes of tangles are most pronounced near the X-point, and the tangles become more complicated as more and more toroidal circuits are traversed; and lobes for closed surfaces become smaller as we move radially inward. Lobes near the X-point are seen experimentally in tokamak [23]. The simple map has much more pronounced tangles [24] compared to the DIII-D. We attribute the differences in tangles of the simple map and the DIII-D to the differences in shape in the two cases.

The average distance $\langle l\rangle$ travelled by field lines on a given surface in a single toroidal turn of the DIII-D varies from about 8 m to 11 m. We set $\langle l\rangle \sim 10$ m. Let $T = T_e = T_i = 50$ eV in the edge region of the DIII-D, where T, T_e and T_i are plasma, electron and ion temperature respectively. Then the thermal velocity of electrons is $v_{Te} \approx 3\times10^6$ m/s, so the time scale for a single toroidal turn for electrons is $\langle l\rangle/v_{Te} \sim 1$ µs. $\langle\Delta r\rangle$ is the average radial displacement of lines from tangles from magnetic perturbations per toroidal circuit of the DIII-D. The separatrix surface has the largest $\langle\Delta r\rangle$ and is roughly about 5 mm to 5 cm. Inner surfaces having smaller displacements.

So the general picture is that in a single turn on the time scale of 1 µs, field lines on the surface travel about 10 m, and are displaced from surface about 5 mm to 5 cm on separatrix surface and suffer smaller displacements on inner surfaces close to separatrix surface. The tangles of the inner surfaces are about the island separatrices. These islands arise due to perturbation, and the internal separatrices only arise when the system is perturbed.

Different types of magnetic perturbations lead to homoclinic tangle of primary separatrix and the attendant lobes that are most pronounced near the X-point. It takes about 1 µs for electrons to complete a single toroidal circuit of the DIII-D. Displacement of electrons on this time scale can generate complicated electric fields in the edge, especially near the X-point. The magnetic field will not respond on this fast time scale; however the structure of flux surfaces becomes sufficiently spatially complicated on an iron gyroradius scale that complicated electric potentials are required to maintain the neutrality of the plasma. The $\mathbf{E}\times\mathbf{B}$ drifts from these complicated electric fields can modify plasma confinement. Smaller lobes for inner surfaces may result in radial gradients in the electric fields which may influence the plasma dynamics in the edge.

The homoclinic tangle is not observed directly in experiments, which observe the deposition of heat and particles. The deposition has clear lobe-like structures, which agree qualitatively with what is expected from homoclinic tangles, but modified by diffusive effects and the electric field, which is required to preserve charge neutrality in the complicated magnetic structures. Such effects are expected to fill in the lobes, which is usually beneficial because it spreads the heat load on the wall and studied in a simple model in Section 3.

The electrons and ions in plasma move rapidly along the magnetic field lines and diffuse slowly across. An important question is how the homoclinic tangle of the magnetic field lines affects the location at which electron and ions diffusing out from the main plasma volume enter the divertor chamber. To represent this non-Hamiltonian effect, we consider a surface 5 cm inside the last good surface on the line from O-point to X-point. Magnetic field lines started on this surface would remain confined forever, but the guiding centers of electrons and ions following the field lines will slowly diffuse across the magnetic field lines until they reach field lines that carry them into the divertor chamber where they are neutralized. One could simulate the diffusion using Monte Carlo methods, but a far more efficient and unprejudiced numerical procedure is to move the particles outward by a small fractional amount $D \ll 1$ after each iteration of the map. Every time a particle crosses the *x*-axis, the radial position of the particle is increased by the factor $(1+RD)$, i.e., $r \to (1+RD)r$, where R is a random number between 0 and 1.

The simulations were started by launching thirty-six thousand particles on the surface 5 cm inside the separatrix with a uniform spacing in poloidal angle and on the $\varphi = 0$ plane. The electron toroidal transit time in DIII-D is of order 1 µs as noted earlier, and the confinement time is about 100 ms, so an electron can make about 10^5 transits while it is confined. The 1000 transits that is used is about right for covering the last 10% of the plasma radius, which is the only thing that is relevant for the formation of lobes in the divertor deposition structure. The strike points of particles are calculated using the continuous analog of the area-preserving DIII-D map [11,12]. The inboard collector plate is located at $x = x_{PLATE} = -0.655$ m. The NCC are mapped to the physical coordinates (x, y) by $x = \sqrt{(2\psi/B_0)}\cos(\theta)$ and $y = \sqrt{(2\psi/B_0)}\sin(\theta)$ [10]. B_0 is the equilibrium field on the magnetic axis of the DIII-D, $B_0 = 1.589$ T [11,12]. We consider the perturbation $(m, n) = (3, 1) + (4, 1)$ with amplitude $\delta = 10^{-4}$. We show the footprints for $D = 10^{-2}, 10^{-3}, 10^{-4}$, and 5×10^{-5} in Fig. 5. For $D \leq 10^{-5}$, no particle strikes the plate. As expected, in the limit as $D \to 0$ the plasma footprint approaches the magnetic footprint, but the effects of the homoclinic tangle remain apparent at $D = 10^{-3}$ though not at 10^{-2}. When the plasma diffusion is sufficiently rapid the effects of the tangle are washed out.

The deposition of plasma heat and particles in the divertor region has the complicated spatial structure that is expected from homoclinic tangles. Although the basic structure of the deposition is controlled by the magnetic field, electric fields and plasma diffusion complicate comparisons and neither the magnetic fields nor the electric fields and diffusive effects are precisely known. Nonetheless, it is important to understand the fundamental features of the deposition patterns of particles that diffuse from the plasma interior into the region in which the magnetic field lines form a homoclinic tangle.

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